

# **MLflow**

## **MLOps**

# MLflow

- Open-source платформа для MLOps
- Разработана компанией Databricks
- Позволяет трекать эксперименты, хранить модели и деплоить их
- Используется в индустрии и research-проектах

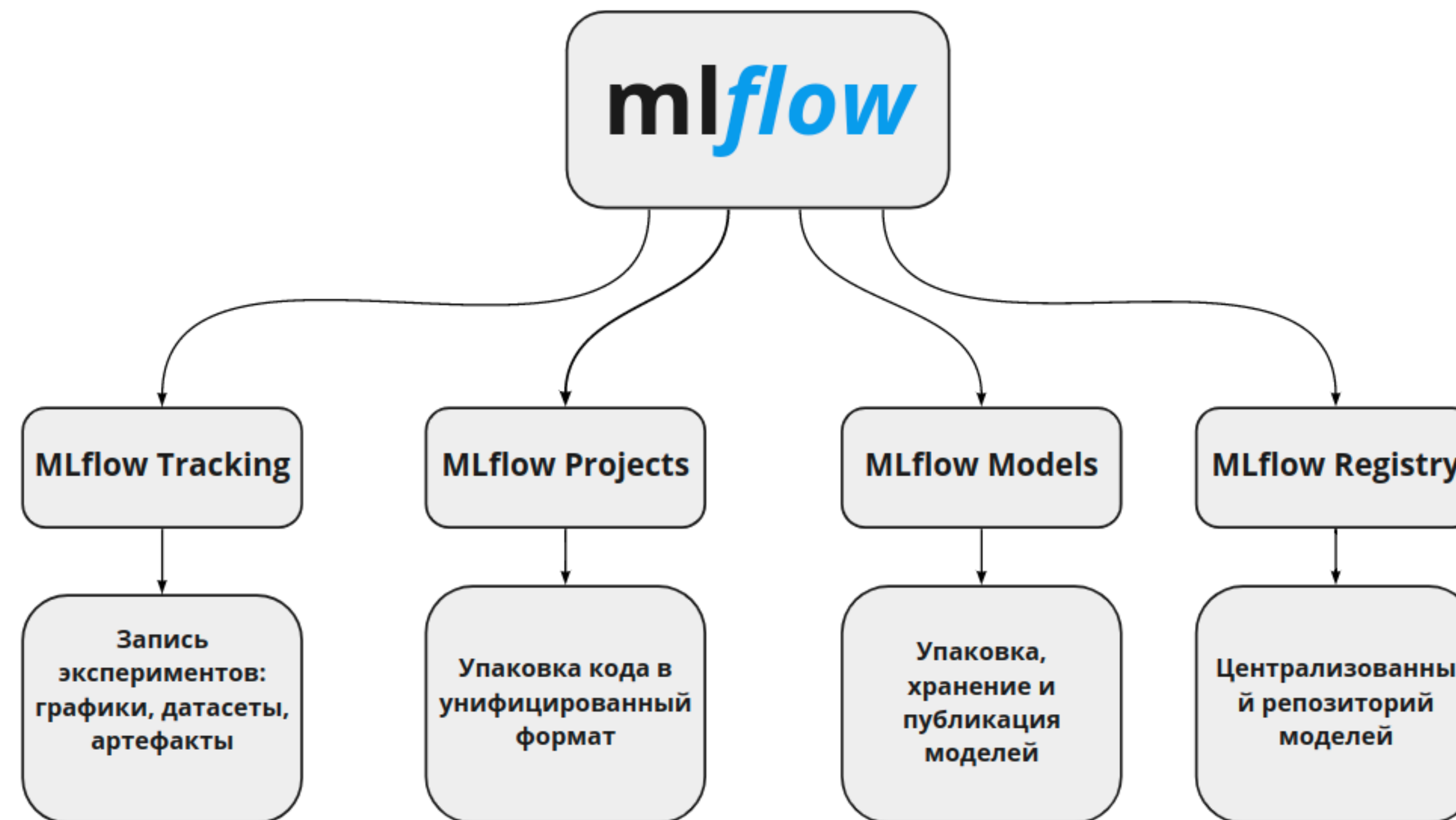
# Проблемы ML-разработки

- Потеря параметров экспериментов
- Сложность воспроизводимости результатов
- Отсутствие контроля версий моделей
- Проблемы совместной работы команды
- Сложный деплой ML-моделей



# MLflow

открытая платформа для управления полным жизненным циклом машинного обучения (ML) и ИИ, разработанная для отслеживания экспериментов, упаковки кода, деплоя моделей и работы с LLM

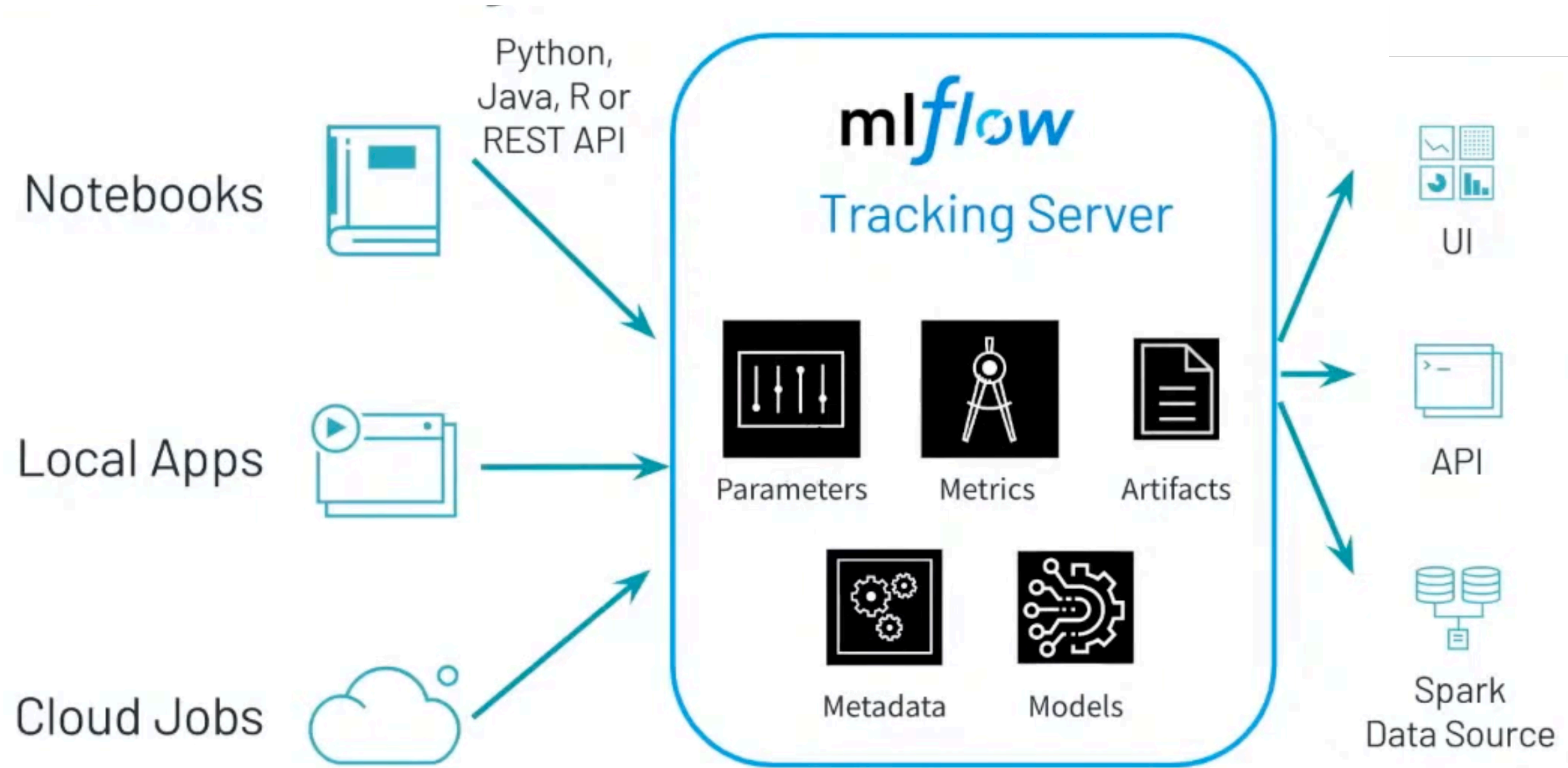




# MLflow + DVC

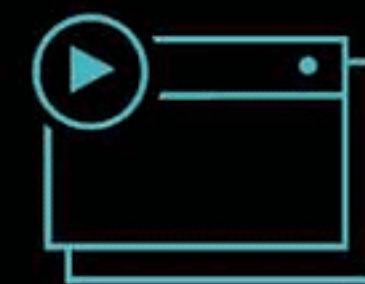
- DVC:
  - версионирование данных и других артефактов
  - воспроизведение экспериментов
- MLflow:
  - трэкинг экспериментов
  - упаковка кода в воспроизводимый формат
  - хранить, управлять и деплоить модели

# MLflow Tracking



# MLflow Projects

## MLflow Projects



Local Execution



Remote Execution



databricks

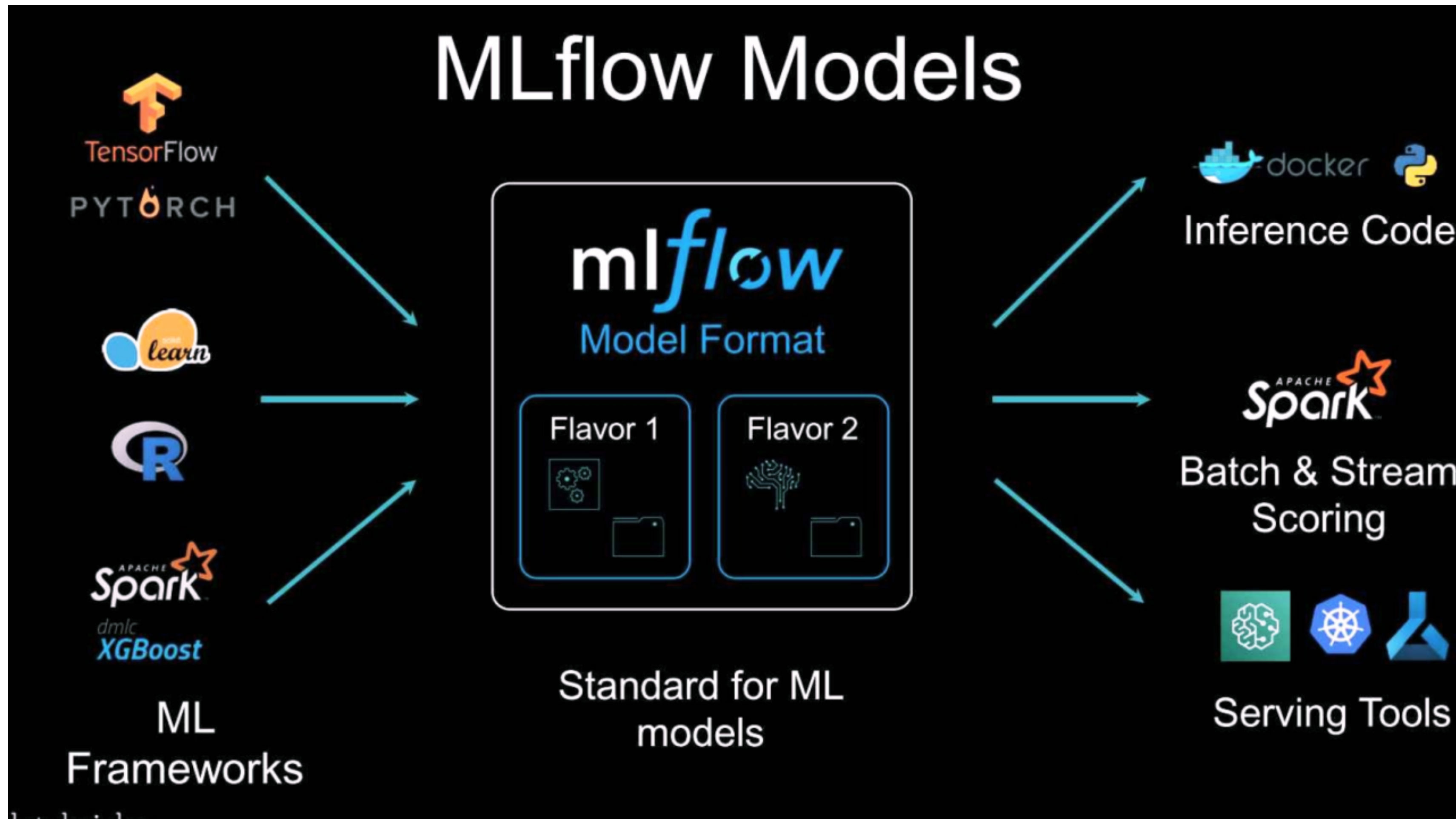


```
my_project/
├── MLproject
│   ├── conda_env: conda.yaml
│   ├── entry_points:
│   │   main:
│   │     parameters:
│   │       training_data: path
│   │       lambda: {type: float, default: 0.1}
│   │     command: python main.py {training_data} {lambda}
│   ├── conda.yaml
│   ├── main.py
│   ├── model.py
│   └── ...
└── ...
```

```
$ mlflow run git://<my_project>
mlflow.run("git://<my_project>", ...)
```



# MLflow Models



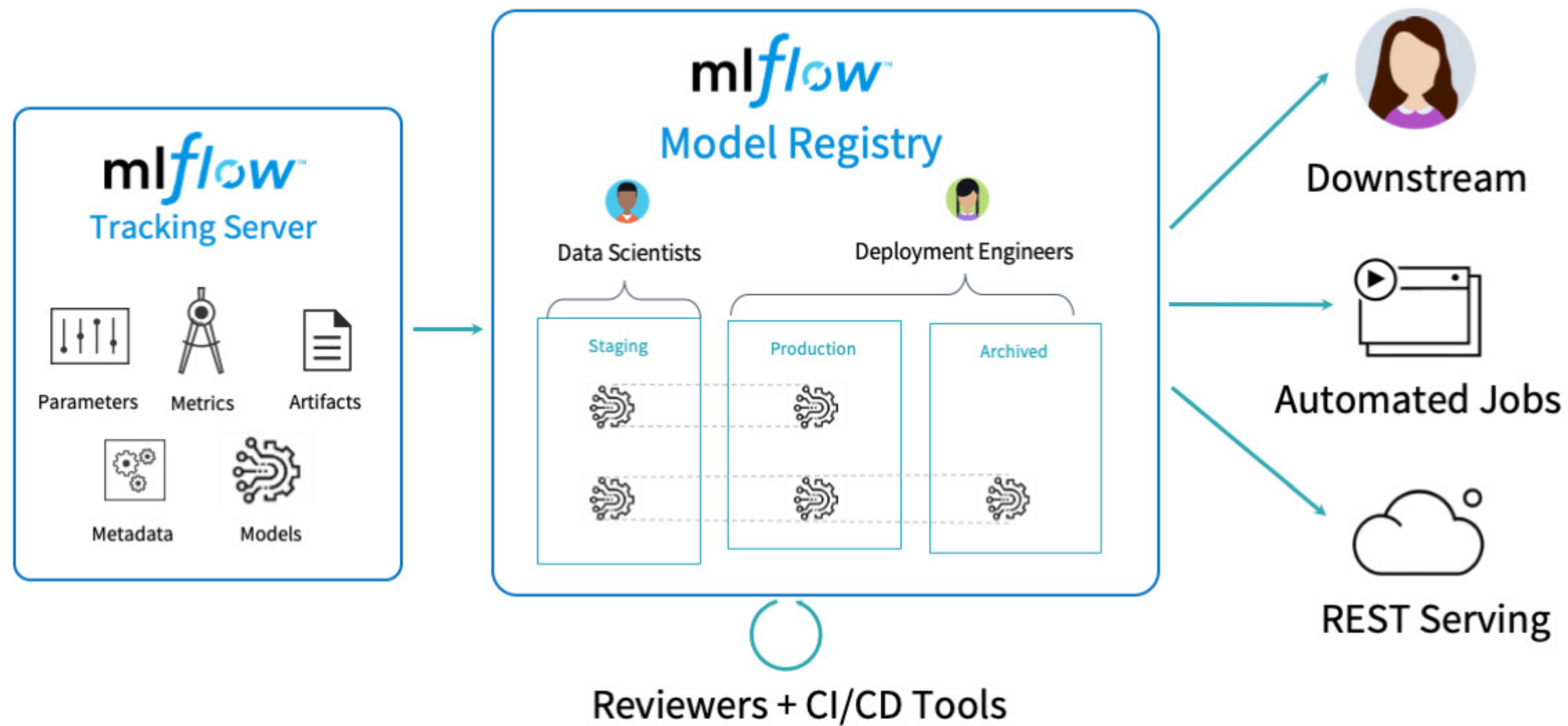
## Example MLflow Model

```
my_model/  
├── MLmodel  
│   ├── run_id: 769915006efd4c4bbd662461  
│   ├── time_created: 2018-06-28T12:34  
│   └── flavors:  
│       ├── tensorflow:  
│       │   ├── saved_model_dir: estimator  
│       │   ├── signature_def_key: predict  
│       └── python_function:  
│           ├── loader_module: mlflow.tensorflow  
└── estimator/  
    ├── saved_model.pb  
    └── variables/  
    ...
```

Usable by tools that understand TensorFlow model format

Usable by any tool that can run Python (Docker, Spark, etc!)

# MLflow Registry



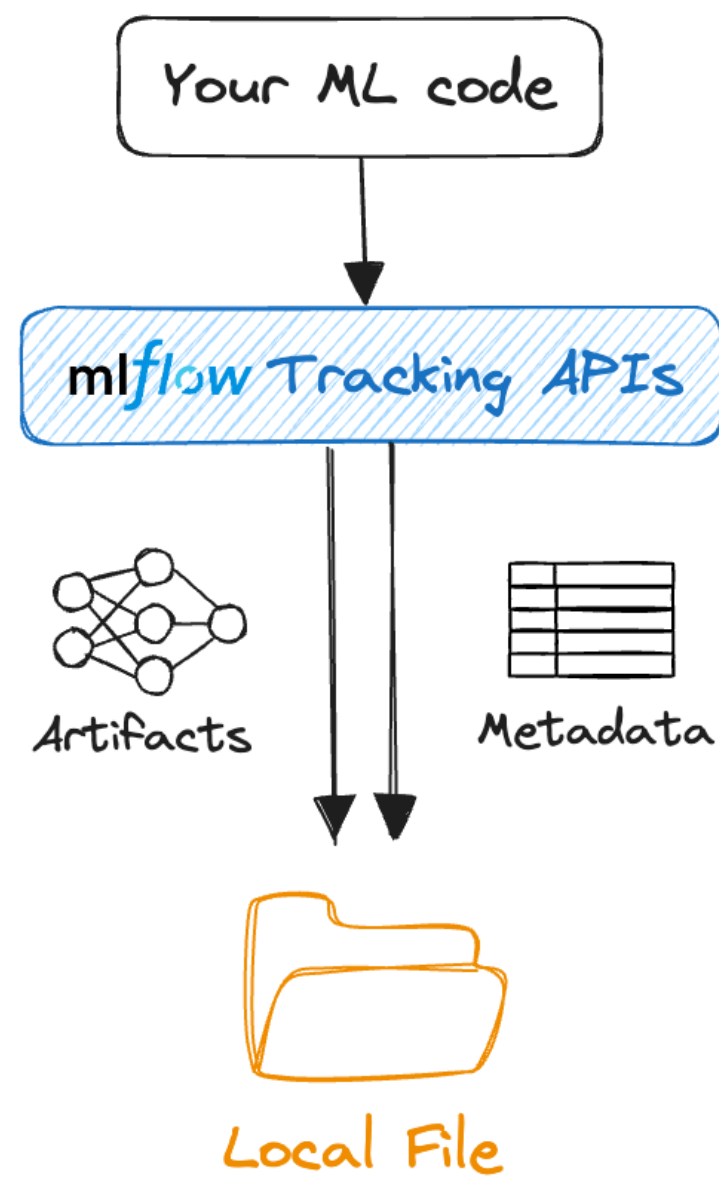
# MLflow установка

- `pip install mlflow/poentry add mlflow`
- `mlflow server —backend-store-uri <PATH>`
- `mlflow ui`

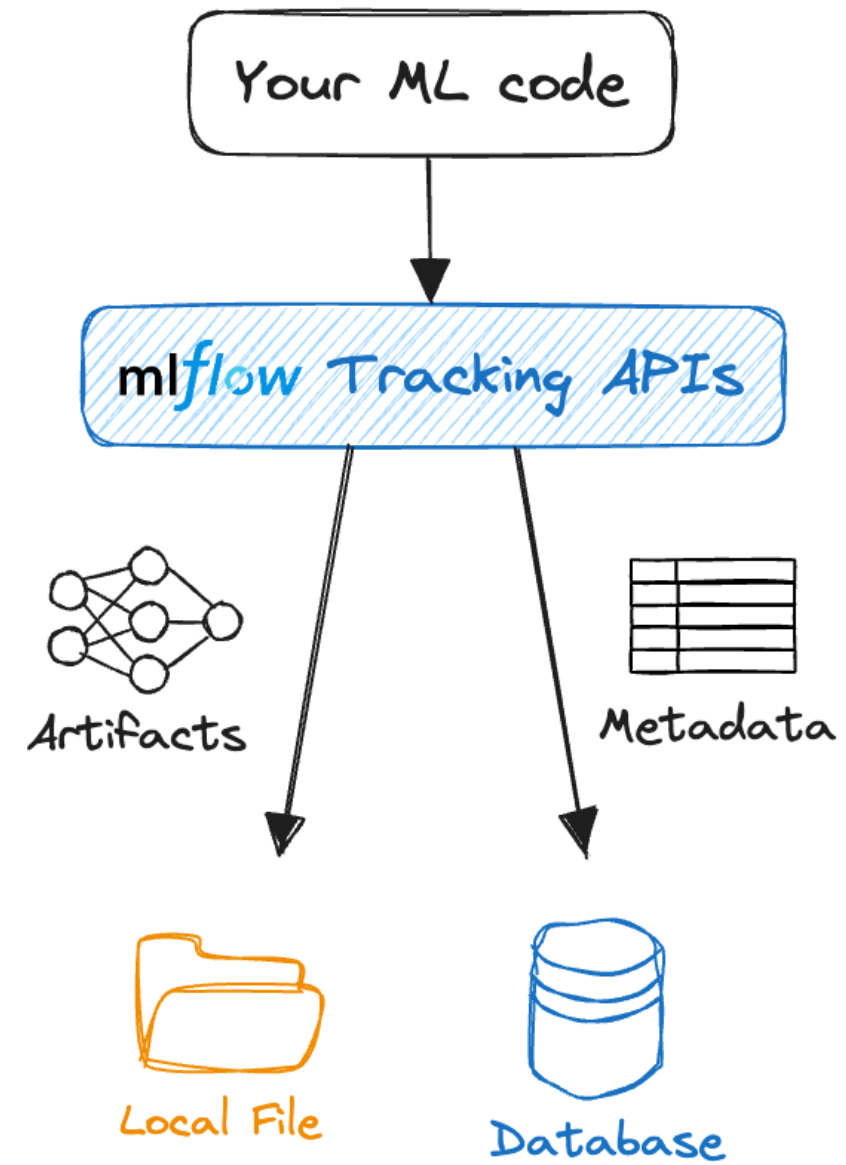


# MLflow

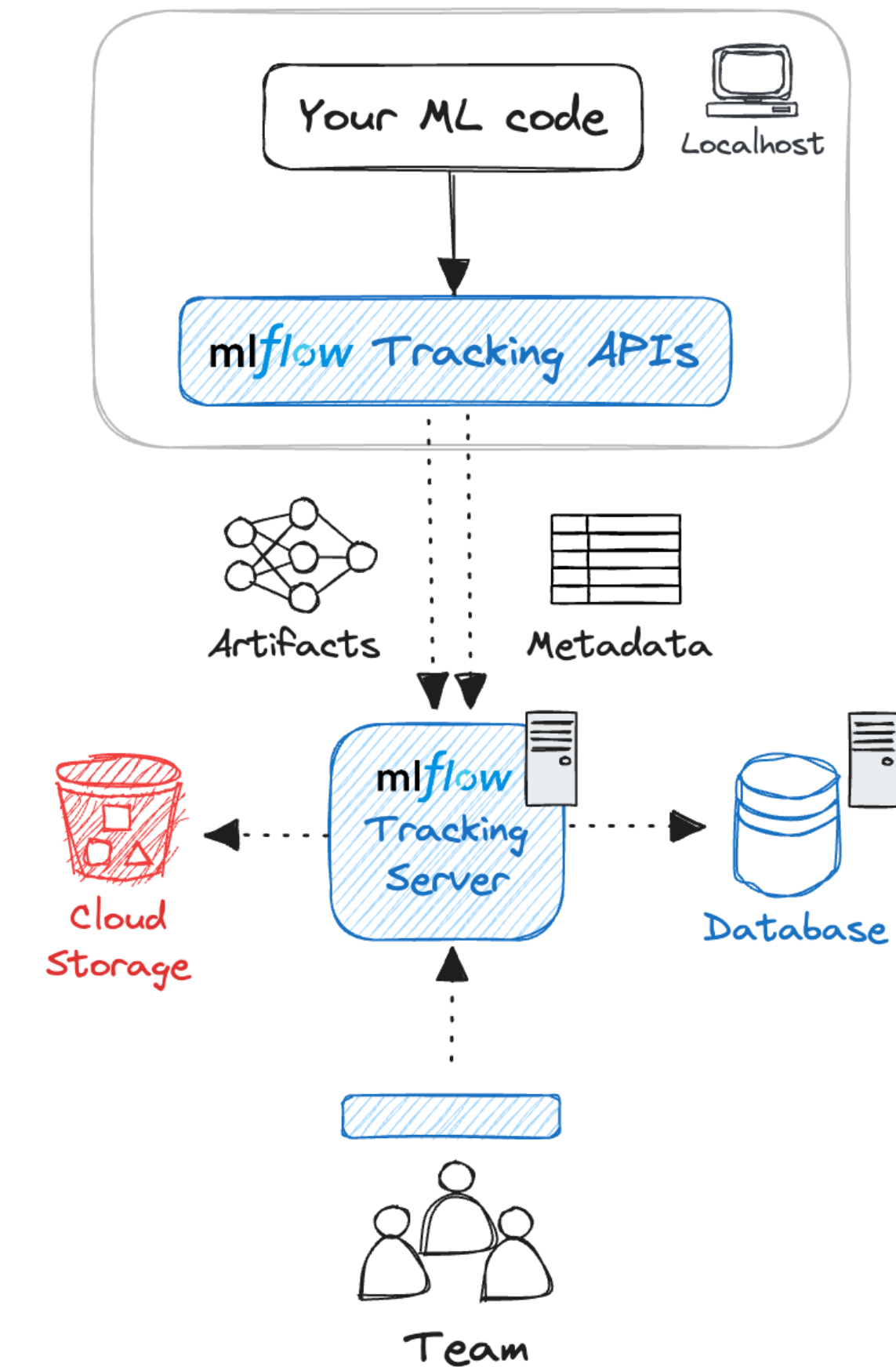
## 1. Localhost (default)



## 2. Localhost w/ various data stores



## 3. Remote Tracking w/ Tracking Server





# А зачем БД и S3?

- БД хранит структурированную информацию о ваших экспериментах:
  - Логирование запусков (Runs): Названия экспериментов, время запуска, длительность, статус
  - Параметры и метрики: гиперпараметры (learning rate, batch size) и метрики качества (accuracy, loss) для каждого шага обучения
  - Теги и комментарии: кто, когда и зачем запускал эксперимент
  - Реестр моделей: информация о версиях моделей, их стадиях (staging, production)
- S3 хранит «тяжёлые» данные, которые генерируются в процессе обучения:
  - Сохраненные модели (артефакты): сериализованные файлы моделей (PyTorch, TensorFlow)
  - Графики и изображения: визуализации, confusion matrix, ROC-кривые
  - Логи обучения: файлы логов в текстовом формате



# Эксперимент

- python mlflow-example.py

mlflow-example.py > ...

```
1  import mlflow
2
3  import mlflow.sklearn
4  from sklearn.datasets import load_diabetes
5  from sklearn.ensemble import RandomForestRegressor
6  from sklearn.model_selection import train_test_split
7
8  mlflow.set_experiment("sklearn_example")
9  mlflow.sklearn.autolog()
10 with mlflow.start_run():
11     db = load_diabetes()
12     X_train, X_test, y_train, y_test = train_test_split(db.data, db.target)
13
14     random_forest = RandomForestRegressor(n_estimators=200, max_depth=6, max_features=10)
15     random_forest.fit(X_train, y_train)
16
17     predictions = random_forest.predict(X_test)
```

# MLflow Experiments

mlflow

3.12.0

GenAI

Model training

←

sklearn\_example

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Tag

▼

| <input type="checkbox"/> | Name            | Time created            | Last modified ↕         | Description | Tags |
|--------------------------|-----------------|-------------------------|-------------------------|-------------|------|
| <input type="checkbox"/> | sklearn_example | 05/10/2026, 07:14:22 PM | 05/10/2026, 07:14:22 PM | -           |      |
| <input type="checkbox"/> | Default         | 05/10/2026, 07:14:22 PM | 05/10/2026, 07:14:22 PM | -           |      |

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# MLflow данные об эксперименте

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upset-hen-926

Overview

Model metrics

System metrics

Traces

Artifacts

Description

No description

Metrics (5)

Search metrics

| Metric                           | Value              | Models |
|----------------------------------|--------------------|--------|
| training_mean_squared_error      | 1206.8643144554574 | model  |
| training_mean_absolute_error     | 29.385767044976223 | model  |
| training_r2_score                | 0.789609326021129  | model  |
| training_root_mean_squared_error | 34.73995271233767  | model  |
| training_score                   | 0.789609326021129  | model  |

Parameters (18)

Search parameters

| Parameter                | Value         |
|--------------------------|---------------|
| bootstrap                | True          |
| ccp_alpha                | 0.0           |
| criterion                | squared_error |
| max_depth                | 6             |
| max_features             | 3             |
| max_leaf_nodes           | None          |
| max_samples              | None          |
| min_impurity_decrease    | 0.0           |
| min_samples_leaf         | 1             |
| min_samples_split        | 2             |
| min_weight_fraction_leaf | 0.0           |

About this run

Created at

05/10/2026, 07:19:06 PM

Created by

leomag

Experiment ID

1

Status

Finished

Run ID

ac49a950132847e6aa0500195d19013c

Duration

2.9s

Source

mlflow-example.py f188396

Registered prompts

—

Datasets

dataset (5698b3ad)

Train

+1

Tags

estimator\_name: RandomForestRegressor

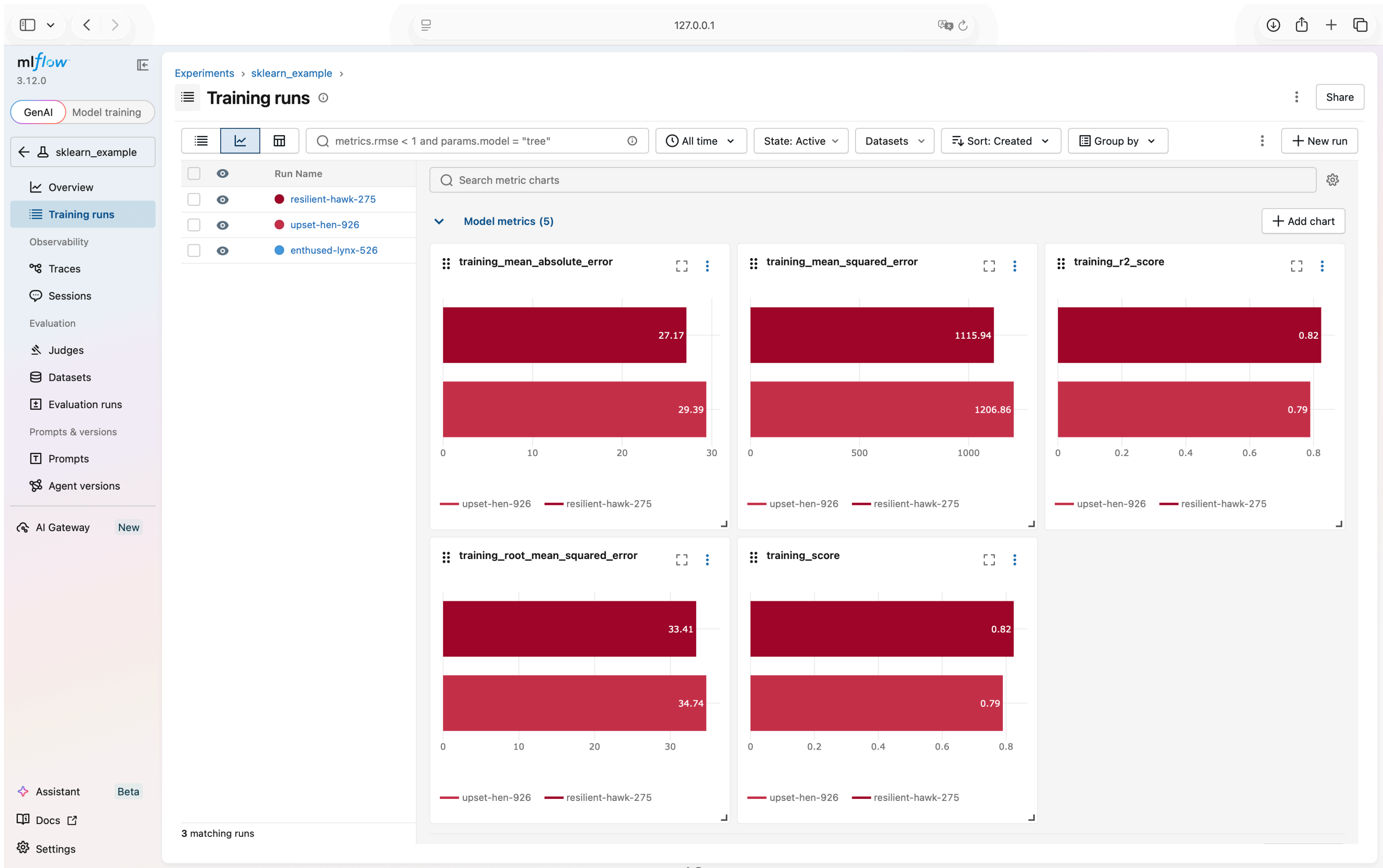
estimator\_class: sklearn.ensemble.\_forest.Ran...

Registered models

None



# MLflow UI



# Логирование и сигнатура модели

- `mlflow server --backend-store-uri sqlite:///mlflow.db --default-artifact-root ./mlruns/artifacts/ --host 127.0.0.1`

```
import mlflow

from mlflow.models import infer_signature
import mlflow.sklearn
import mlflow.sklearn
from sklearn.datasets import load_diabetes
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split

mlflow.set_tracking_uri("http://127.0.0.1:5000")
mlflow.set_experiment("sklearn_example")
mlflow.sklearn.autolog()
with mlflow.start_run():
    db = load_diabetes()
    X_train, X_test, y_train, y_test = train_test_split(db.data, db.target)

    random_forest = RandomForestRegressor(n_estimators=200, max_depth=6, max_features=10)
    random_forest.fit(X_train, y_train)

    predictions = random_forest.predict(X_test)

    signature = infer_signature(X_train, predictions)

    mlflow.sklearn.log_model(random_forest, "diabetes_random_forest", signature=signature)
```



# Регистрация модели

mlflow3.12.0

GenAIModel training

←sklearn\_example

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diabetes\_random\_forest

Not registered

Register model

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Description

No description

Metrics (5)

Search metrics

| Metric                           | Dataset | Source run       | Value              |
|----------------------------------|---------|------------------|--------------------|
| training_mean_squared_error      | -       | powerful-dog-483 | 1159.239136708104  |
| training_mean_absolute_error     | -       | powerful-dog-483 | 27.952155738422... |
| training_r2_score                | -       | powerful-dog-483 | 0.7919717005994... |
| training_root_mean_squared_error | -       | powerful-dog-483 | 34.0476010418958   |
| training_score                   | -       | powerful-dog-483 | 0.7919717005994... |

Parameters (18)

Search parameters

| Parameter                | Value         |
|--------------------------|---------------|
| bootstrap                | True          |
| ccp_alpha                | 0.0           |
| criterion                | squared_error |
| max_depth                | 6             |
| max_features             | 10            |
| max_leaf_nodes           | None          |
| max_samples              | None          |
| min_impurity_decrease    | 0.0           |
| min_samples_leaf         | 1             |
| min_samples_split        | 2             |
| min_weight_fraction_leaf | 0.0           |

About this logged model

Created at

2 minutes ago

Created by

leomag

Status

Ready

Model ID

m-9682ce8545ef4b279d...

Source run

powerful-dog-483

Source run ID

9dc6e2170a0e4a20a174f...

Logged from

mlflow-example.py

f188396

Datasets used

None

Model versions

None

# Зарегистрированная модель

Частный доступ

127.0.0.1

mlflow

3.12.0

GenAI

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OverviewModel metricsSystem metricsTracesArtifacts

| Parameter                | Value         |
|--------------------------|---------------|
| bootstrap                | True          |
| ccp_alpha                | 0.0           |
| criterion                | squared_error |
| max_depth                | 6             |
| max_features             | 10            |
| max_leaf_nodes           | None          |
| max_samples              | None          |
| min_impurity_decrease    | 0.0           |
| min_samples_leaf         | 1             |
| min_samples_split        | 2             |
| min_weight_fraction_leaf | 0.0           |
| monotonic_cst            | None          |
| n_estimators             | 200           |
| n_jobs                   | None          |

Registered models

sklearn-model v1

Logged models (2)

| Model attributes |      |                 |        |                |                   |                   |
|------------------|------|-----------------|--------|----------------|-------------------|-------------------|
| Type             | Step | Model name      | Status | Created        | Registered models | Dataset           |
| Output           | 0    | diabetes_random | Ready  | 11 minutes ago | sklearn-model v1  | -                 |
| Output           | 0    | model           | Ready  | 11 minutes ago | -                 | dataset (#9567cd) |

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# Зарегистрированная модель

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3.12.0

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| Name          | Latest version | Aliased versions | Created by | Last modified         | Tags |
|---------------|----------------|------------------|------------|-----------------------|------|
| sklearn-model | Version 1      |                  |            | 05/10/2026, 07:57:... | —    |

New model registry UI

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# Развернуть в контейнерах на сервере

## docker-compose.yml

- Portainer - веб-интерфейсу для управления Docker-контейнерами, Kubernetes и Docker Swarm

```
portainer:
  image: portainer/portainer-ce:latest
  container_name: portainer
  restart: always
  security_opt:
    - no-new-privileges:true
  volumes:
    - /etc/localtime:/etc/localtime:ro
    - /var/run/docker.sock:/var/run/docker.sock:ro
    - portainer_data:/data
  ports:
    - 9443:9443
    - 8000:8000
```

# Развернуть в контейнерах на сервере

## docker-compose.yml

```
minio:
  image: quay.io/minio/minio:latest
  container_name: minio
  restart: always
  ports:
    - "9000:9000"    # API Access
    - "9001:9001"    # Web Console Access
  environment:
    MINIO_ROOT_USER: ${MINIO_ROOT_USER}
    MINIO_ROOT_PASSWORD: ${MINIO_ROOT_PASSWORD}
  healthcheck:
    test: ["CMD", "curl", "-f", "http://localhost:9000/minio/health/live"]
    interval: 10s
    timeout: 5s
    retries: 5
  volumes:
    - minio_data:/data
  command: server --console-address ":9001" /data
  networks:
    - minio-network
  depends_on:
    db:
      condition: service_healthy
```

```
db:
  image: postgres:15
  container_name: postgres_db
  restart: always
  environment:
    POSTGRES_USER: ${POSTGRES_USER}
    POSTGRES_PASSWORD: ${POSTGRES_PASSWORD}
    POSTGRES_DB: ${POSTGRES_DB}
    PG_DATA: data/postgres
  ports:
    - "5432:5432"
  healthcheck:
    test: ["CMD-SHELL", "pg_isready -U ${POSTGRES_USER}"]
    interval: 10s
    timeout: 5s
    retries: 5
  volumes:
    - postgres_data:/data/postgres
  networks:
    - postgres-network
  > Run Service
pgadmin:
  image: dpage/pgadmin4
  container_name: pgadmin_container
  restart: always
  environment:
    PGADMIN_DEFAULT_EMAIL: ${PG_ADMIN_EMAIL}
    PGADMIN_DEFAULT_PASSWORD: ${PG_ADMIN_PASSWORD}
  ports:
    - "5050:80"
  depends_on:
    db:
      condition: service_healthy
  networks:
    - postgres-network
```



# Развернуть в контейнерах на сервере

## `docker-compose.yml`

```
mlflow:
  build:
    context: .
    dockerfile: Dockerfile
  container_name: mlflow_container
  restart: always
  ports:
    - "5001:5000"
  networks:
    - minio-network
    - postgres-network
  environment:

    MLFLOW_S3_ENDPOINT_URL: ${MLFLOW_S3_ENDPOINT_URL}
    MLFLOW_S3_ACCESS_KEY_ID: ${MINIO_ROOT_USER}
    MLFLOW_S3_SECRET_ACCESS_KEY: ${MINIO_ROOT_PASSWORD}

  command: mlflow server --backend-store-uri postgresql://${POSTGRES_USER}:${POSTGRES_PASSWORD}@db/${POSTGRES_DB} --default-artifact-root s3://${BUCKET_NAME} --host 0.0.0.0

  depends_on:
    db:
      condition: service_healthy
    minio:
      condition: service_healthy
```

# Развернуть в контейнерах на сервере

## docker-compose.yml

```
volumes:
  portainer_data:
  postgres_data:
  minio_data:

networks:
  minio-network:
    driver: bridge
  postgres-network:
    driver: bridge
```

```
.env
1 PG_ADMIN_EMAIL=user@itmo.ru
2 PG_ADMIN_PASSWORD=adminpassword
3 POSTGRES_PASSWORD=password123
4 POSTGRES_USER=admin
5 POSTGRES_DB=mydatabase
6 MINIO_ROOT_USER=minioadmin
7 MINIO_ROOT_PASSWORD=minioadmin
8 MLFLOW_S3_ENDPOINT_URL=http://localhost:9000
9 BUCKET_NAME=mlflow
```

# Добавить код

```
load_dotenv()

os.environ["AWS_ACCESS_KEY_ID"] = os.getenv("MINIO_ROOT_USER")
os.environ["AWS_SECRET_ACCESS_KEY"] = os.getenv("MINIO_ROOT_PASSWORD")
os.environ["MLFLOW_S3_ENDPOINT_URL"] = os.getenv("MLFLOW_S3_ENDPOINT_URL")

ARTIFACT_LOCATION = "s3://" + os.getenv("BUCKET_NAME")

mlflow.set_tracking_uri("http://localhost:5001")
if mlflow.get_experiment_by_name("sklearn_example2") is None:
    mlflow.create_experiment("sklearn_example2",
                            artifact_location=ARTIFACT_LOCATION)
mlflow.set_experiment("sklearn_example2")
mlflow.sklearn.autolog()
with mlflow.start_run():
    db = load_diabetes()
    X_train, X_test, y_train, y_test = train_test_split(db.data, db.target)

    random_forest = RandomForestRegressor(n_estimators=200,
                                         max_depth=6,
                                         max_features=10)
    random_forest.fit(X_train, y_train)

    predictions = random_forest.predict(X_test)
```

```
def ensure_s3_bucket_for_artifact_uri(artifact_uri: str) -> None:
    """Create the S3/MinIO bucket if missing (PutObject fails with NoSuchBucket otherwise)."""
    parsed = urlparse(artifact_uri)
    if parsed.scheme != "s3" or not parsed.netloc:
        return
    bucket = parsed.netloc
    client = boto3.client(
        "s3",
        endpoint_url=os.environ["MLFLOW_S3_ENDPOINT_URL"],
        aws_access_key_id=os.environ["AWS_ACCESS_KEY_ID"],
        aws_secret_access_key=os.environ["AWS_SECRET_ACCESS_KEY"],
    )
    try:
        client.create_bucket(Bucket=bucket)
    except ClientError as e:
        code = e.response.get("Error", {}).get("Code", "")
        if code not in ("BucketAlreadyOwnedByYou", "BucketAlreadyExists"):
            raise
```



- <https://habr.com/ru/companies/ods/articles/828458/>
- <https://www.youtube.com/watch?v=rfEE4Yc4gXg>